

# Informatics

**Informatics** is a field of study which deals with getting, processing and using information. Information can be saved, **displayed**, **transmitted** through a **transmission** medium, selected, sorted, etc. For these operations we use information technologies, primarily computers. Among the typical sub-disciplines of informatics belong computer technology, robotics, cybernetics, artificial intelligence, computer simulations, telecommunications, etc.

**Computers** are digital machines, so it is necessary to code the information into digital form. This means it is possible to write the information using ones and zeroes. The basic unit of information is **1 bit**. A sequence of 8 bits makes **1 Byte**. Bits and Bytes are small units, therefore we use their multiples, which are a kilobyte (1 kB = 1024 B), a megabyte (1 MB = 1024 kB), a gigabyte (1 GB = 1024 MB) and today also a terabyte (1 TB = 1024 GB).

Nowadays it is important to put emphasis on **data security**. It can be damaged, lost or **misused**. This is a list of **safety precautions**, which should prevent the data from being **debased**:

- physically secure hardware against stealing
- protect the computer when turned on (password, identification card, fingerprint)
- protect software against malware (viruses, worms, Trojan horses, ...)
- code the data
- set user accounts and access rights in the operating system
- use a standby supply UPS (in case of blackouts) = uninterruptible power supply
- work safely and carefully with a computer
- **back up data** regularly

## Vocabulary

|                           |                              |
|---------------------------|------------------------------|
| <i>display</i>            | <i>zobrazit</i>              |
| <i>transmit</i>           | <i>přenést</i>               |
| <i>transmission</i>       | <i>přenos, přenosový</i>     |
| <i>1 Byte /bait/</i>      |                              |
| <i>1 bit /bit/</i>        |                              |
| <i>misuse</i>             | <i>zneužít</i>               |
| <i>safety precautions</i> | <i>bezpečnostní opatření</i> |
| <i>debase</i>             | <i>znehodnotit</i>           |
| <i>back up data</i>       | <i>zálohovat data</i>        |

- What is informatics?
  - Basic unit of information and its multiples
  - Safety precautions which prevent the data from being debased and / or misused
  - Other
- ② What do you do to protect your data?
  - ② Do you open unknown e-mails or files? Why? Why not? Can it be dangerous?
  - ② Do you use antivirus programs?